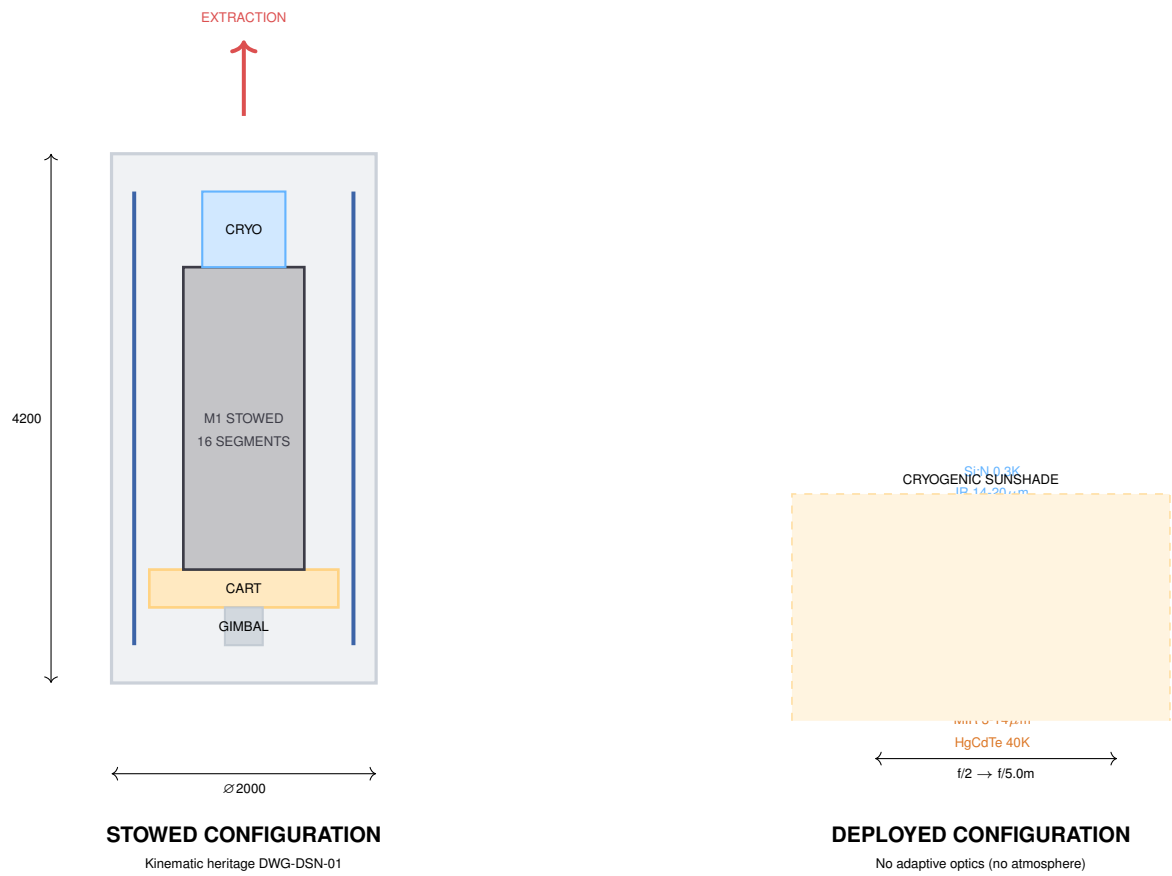


DWG-MIR-01: MIR/IR TELESCOPE — NEO PLANETARY DEFENSE

2.5m Segmented Aperture — Dual Dichroic Channel — TUG Early Warning



MIR/IR TELESCOPE TECHNICAL SPECIFICATIONS

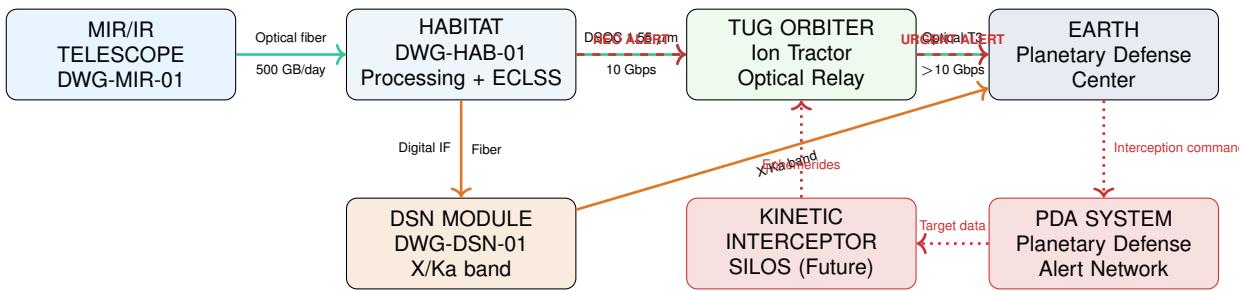
OPTICS AND STRUCTURE			MIR CHANNEL (3–14 μ m)		
Primary aperture	2.5 m (16 CFRP segments)		Detector	HgCdTe 2048 × 512	
Focal ratio	f/2 (primary)		Sampling	0.5"/px (Nyquist @10 μ m)	
Surface precision	< 20 nm RMS		FOV	0.5 deg ²	
Optical bench	Invar 36		Spectral resolution	R ~ 100	
Adaptive optics	Not included (no atmosphere)		Detector temp.	40 K (Pulse Tube)	
IR CHANNEL (14–20 μ m)			COMMUNICATIONS AND ALERT		
Detector	Si:N 256 × 256 bolometers		Surface link	Optical fiber → Habitat	
Sampling	1.0"/px (Nyquist @18 μ m)		Uplink	DSOC 1.55 μ m → TUG	
FOV	0.3 deg ²		Data rate	500 GB/day → 10 Gbps	
Spectral resolution	R ~ 50		Early warning	Absolute TUG priority	
Detector temp.	< 0.3 K (³ He/ ⁴ He dilution)		Silo interface	PDA protocol (future)	

MISSION:	ASTRA / L-RAIL (NASA/COPAG)	REV: A
DRAWING:	DWG-MIR-01 — MIR/IR TELESCOPE	DATE: 2026-08-11
TITLE:	2.5m NEO Telescope — IR Surveillance + Early Warning	SHEET 1/2
SCALE: 1:25	UNITS: mm	FORMAT: A1 L

DWG-MIR-01: COMMUNICATIONS, EARLY WARNING AND DEPLOYMENT

TUG Orbiter Link — Kinetic Interception System Interface — Operational Sequence

COMMUNICATIONS ARCHITECTURE AND NEO EARLY WARNING



EARLY WARNING PROTOCOL (PDA-Link)

- When the telescope detects a NEO with impact probability $> 1\%$ (Torino scale ≥ 1), the priority alert protocol is activated:
- Detection and astrometry:** The MIR channel measures position, magnitude and motion rate of the object. 1000 s exposures in TDI mode.
 - Initial orbit determination:** The habitat processor (FPGA + rad-hard CPU) calculates preliminary orbital elements in < 5 min.
 - Threat classification:** If orbital uncertainty crosses the Earth impact corridor, a PDA-ALERT message is generated.
 - Priority transmission:** The PDA-ALERT message has absolute priority on the DSOC link to the TUG. Any routine scientific data traffic is interrupted.
 - TUG → Earth relay:** The TUG retransmits the alert to the optical ground station (TG) with latency < 1.5 s (lunar distance).
 - Confirmation and tracking:** Continuous follow-up campaign with 18 s cadence (burst mode) begins to refine the orbit.

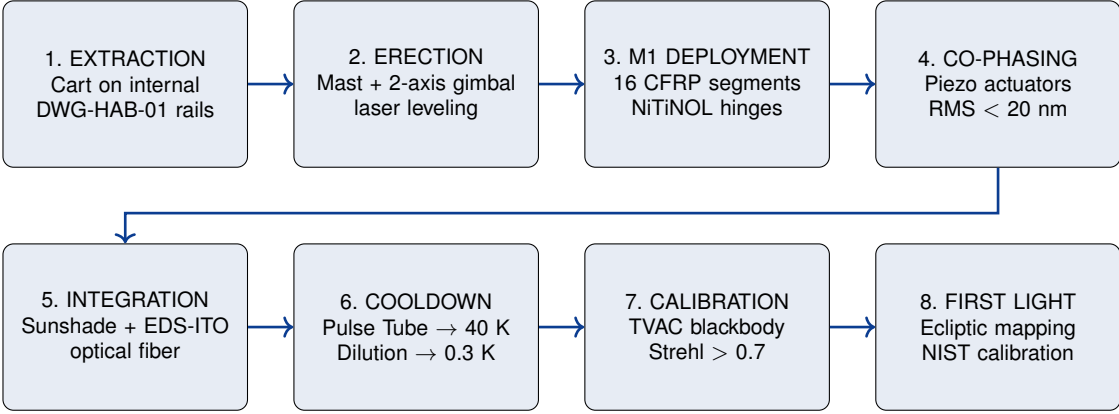
Total detection → alert latency on Earth: < 10 minutes (including processing and transmission).

Redundancy: If the optical TUG link is interrupted (eclipse or maintenance), the DSN module transmits the alert in X-band at 1 kbps as an emergency channel.

KINETIC INTERCEPTION SYSTEM INTERFACE

- The L-RAIL architecture contemplates a future phase of active planetary defense with kinetic interceptors based on the Moon. The MIR/IR telescope is the primary sensor of this system:
- Lunar advantage:** The low escape velocity ($v_{esc} = 2.38$ km/s) drastically reduces the ΔV needed to intercept NEOs on blind trajectories from Earth.
 - Target data:** The telescope provides high-precision ephemerides ($\sigma_{pos} < 0.5$ arcsec), thermal light curve (size/albedo), and taxonomy (C/S/M) for interceptor design.
 - Data protocol:** CCSDS 734.2-B-1 (DTN Bundle Protocol) format with PDA-TARGET encapsulation. Includes: state vector (6-DOF), covariance matrix, shape model, and interception window.
 - Link to silos:** Target data is transmitted via TUG → Earth → Command Center → Lunar silos (when operational). In future autonomous mode, the TUG can retransmit directly to silos with encrypted command.
 - Post-launch tracking:** After interceptor launch, the telescope performs astrometric tracking of the object + interceptor for closed-loop trajectory correction.
- Phase C note:** At this stage, only the data interface and PDA-TARGET protocol are implemented. The silos and interceptor system are Phase D/E elements (post-CDR). The telescope must be designed to be compatible without modifications.
- Applicable standards:** NASA-STD-8739.8 (software), CCSDS 734.2-B-1 (DTN), MIL-STD-461 (EMI), ISO 11227 (debris).

OPERATIONAL DEPLOYMENT SEQUENCE (PHASE C)



DETAILED MASS BUDGET (DWG-MIR-01 — PHASE C)

ID	Component	Mass (kg)	Accumulated (kg)
MIR-001	Primary mirror M1 (16 CFRP segments + actuators)	85.0	85.0
MIR-002	Secondary mirror M2 + support (CFRP struts)	12.0	97.0
MIR-003	ZnSe dichroic splitter + reimaging optics	8.0	105.0
MIR-004	Invar 36 optical bench + Kevlar harnesses	35.0	140.0
MIR-005	MIR camera: HgCdTe 2048×512 + electronics	15.0	155.0
MIR-006	IR camera: Si:N 256×256 + multiplexer	12.0	167.0
MIR-007	Pulse Tube cryocooler (2-stage) + compressor	30.0	197.0
MIR-008	³ He/ ⁴ He dilution cryostat (closed cycle)	22.0	219.0
MIR-009	Cryogenic multilayer sunshade (MLI + V-grooves)	18.0	237.0
MIR-010	EDS-ITO system (electrodes + HV supply)	6.0	243.0
MIR-011	2-axis gimbal + mast + extraction cart	28.0	271.0
MIR-012	Optical fiber wiring + control electronics	10.0	281.0
MIR-013	Surface DSOC link (1.55 μm optical terminal)	9.0	290.0
Subtotal		290.0	
System margin (15 %)		43.5	
TOTAL DWG-MIR-01		333.5	

Mass balance note: With DWG-HAB-01 (460 kg), DWG-DSN-01 (205 kg), VLBI Phase A (515 kg), xenon (8.3 kg), NEXT-C (100 kg), Tug structure (450 kg), and DWG-MIR-01 (333.5 kg), the accumulated mass is ~2072 kg. The Tug budget is 2869 kg, leaving ~797 kg for the VADS, the MnN reactor, and the extraction rover. The mass is compatible with the 2-Tug architecture.

Savings note from absence of AO: Not including adaptive optics saves ~50 kg and ~80 W, directly redirected to the dilution cryostat margin and the early warning DSOC terminal.

Next milestones (CDR → Phase D): Mirror segment prototype with EDS-ITO in cryogenic TVAC; 30-day continuous dilution cryostat demonstration; validation of surface-TUG DSOC link at 10 Gbps; integration of PDA-TARGET protocol with NEO campaign simulation.

NORMATIVE NOTES AND PHASE C REFERENCES

- ISO 10110:** Optical quality specification. Strehl > 0.7 at $10\text{ }\mu\text{m}$. Verification with cryogenic interferometer.
- NIST SP 250-94:** Absolute radiometric calibration. Traceable cavity blackbody. Uncertainty $< 5\%$.
- NRC (2010) — Defending Planet Earth:** Sensitivity requirements for detection of PHAs > 140 m. Justification of MIR/IR band.
- Mainzer et al. (2015) — NEOWISE:** TDI mapping operation concept. Direct heritage of the MIR channel.
- NASA TM-2010-216213 + Calle et al. (2016):** EDS-ITO system. Efficiency $> 95\%$ without optical degradation.
- CCSDS 734.2-B-1 (DTN):** Disruption-tolerant communication protocol. PDA-TARGET format for interception data.
- MIL-STD-461:** Electromagnetic interference control. Compatibility between EDS-ITO, cryocooler and DSOC link.
- NASA-STD-8739.8:** Flight software. Validation of orbit determination algorithm and alert protocol.
- ISO 11227:** Debris mitigation. Zero particles released during deployment and EDS operation.
- Compatibility with DWG-HAB-01:** Mechanical interface (HAB-005 rails), thermal (shared HAB-010 loop), and data (HAB-007 fiber).

MISSION:	ASTRA / L-RAIL (NASA/COPAG)	REV: A
DRAWING:	DWG-MIR-01 — MIR/IR TELESCOPE	DATE: 2026-08-11
TITLE:	2.5m NEO Telescope — IR Surveillance + Early Warning	SHEET 1/2
SCALE: 1:25	UNITS: mm	FORMAT: A1 L